

Jin Ma

PH.D. CANDIDATE · COMPUTER SCIENCE

Clemson University, Clemson, SC 29634

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Profile

Jin Ma is a Ph.D. candidate in Computer Science at Clemson University specializing in trustworthy AI and cybersecurity, with a focus on computer vision and machine learning security. His research examines vulnerabilities in vision-based perception systems, generative-model-based defenses against adversarial attacks, and applications of large vision-language models (LVLMs) to online content safety and misinformation mitigation. His research is informed by prior industry R&D experience and interdisciplinary collaborations across security, multimedia, and intelligent systems.

Research interests: Trustworthy AI, Robust Vision-Based Perception Systems, Multimodal Web Safety

Education

Clemson University

PH.D. COMPUTER SCIENCE

• Advisor: Dr. Long Cheng; GPA: 4.0 / 4.0.

Clemson, SC, USA

Aug. 2023 – Aug. 2027 (expected)

Cleveland State University

PH.D. STUDIES IN COMPUTER SCIENCE (TRANSFERRED TO CLEMSON)

• Advisor: Dr. Hongkai Yu; GPA: 4.0 / 4.0.

Cleveland, OH, USA

Aug. 2021 – Jul. 2023

Xi'an Jiaotong University

M.E. SOFTWARE ENGINEERING

• Advisor: Dr. Shanmin Pang.

Xi'an, Shaanxi, China

Sep. 2016 – Jun. 2019

Xi'an Jiaotong University

B.S. INFORMATION AND COMPUTATIONAL SCIENCE

Xi'an, Shaanxi, China

Sep. 2012 – Jun. 2016

Publications

Citation Summary (September 2026): 760+ citations · *h*-index: 11 · *i*10-index: 11 (* Co-first author; † Student mentee)

REFEREED JOURNAL ARTICLES

Ma, J.*, Enan, A.*, Cheng, L., & Chowdhury, M. (2026). Understanding the risks of asphalt art to the reliability of vision-based perception systems. *Transportation Research Record (TRR)*. [In press].

Aftabi, N., Samaha, P., **Ma, J.**, Cheng, L., Harik, R., & Li, D. (2026). ViSTR-GP: Online cyberattack detection via vision-to-state tensor regression and Gaussian processes in automated robotic operations. *Journal of Manufacturing Systems*. [Accepted].

Li, J., Xu, R., Liu, X., **Ma, J.**, Li, B., Zou, Q., Ma, J., & Yu, H. (2024). Domain adaptation based object detection for autonomous driving in foggy and rainy weather. *IEEE Transactions on Intelligent Vehicles (T-IV)*, 10(2), 900–911.

Li, J., Xu, R., Liu, X., **Ma, J.**, Chi, Z., Ma, J., & Yu, H. (2023). Learning for vehicle-to-vehicle cooperative perception under lossy communication. *IEEE Transactions on Intelligent Vehicles (T-IV)*, 8(4), 2650–2660.

Liu, X., Li, J., **Ma, J.**, Sun, H., Xu, Z., Zhang, T., & Yu, H. (2023). Deep transfer learning for intelligent vehicle perception: A survey. *Green Energy and Intelligent Transportation*, 2(5), 100125.

Sun, H., **Ma, J.**, Guo, Q., Zou, Q., Song, S., Lin, Y., & Yu, H. (2023). Coarse-to-fine task-driven inpainting for geoscience images. *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, 33(12), 7170–7182.

Pang, S., **Ma, J.**, Xue, J., Zhu, J., & Ordonez, V. (2018). Deep feature aggregation and image re-ranking with heat diffusion for image retrieval. *IEEE Transactions on Multimedia (T-MM)*, 21(6), 1513–1523.

Pang, S., **Ma, J.**, Zhu, J., Xue, J., & Tian, Q. (2018). Improving object retrieval quality by integration of similarity propagation and query expansion. *IEEE Transactions on Multimedia (T-MM)*, 21(3), 760–770.

REFEREED CONFERENCE PROCEEDINGS

Rudharraju, A.[†], Cheng, E., Tong, W., & **Ma, J.** (2026). Can AI be a doctor? Benchmarking large vision-language models on ophthalmic tasks. *IEEE BIBM 2026 Undergraduate and High School Symposium (UGHS 2026)*. [Accepted].

Yan, J., Liao, S., **Ma, J.**, Aldeen, M. S., Kumar, S., & Cheng, L. (2025). No way to sign out? Unpacking non-compliance with Google Play's app account deletion requirements. *34th USENIX Security Symposium (USENIX Security 25)*, 3277–3296.

Wang, F., Sun, F., Fan, M., Zhou, J., **Ma, J.**, Chen, C., Shu, J., & Zhang, L. Y. (2025). FLAME: Flexible and lightweight biometric authentication scheme in malicious environments. *41st Annual Computer Security Applications Conference (ACSAC 2025)*, 411–424.

Aldeen, M. S., MohajerAnsari, P., **Ma, J.**, Chowdhury, M., Cheng, L., & Pesé, M. D. (2024). An initial exploration of employing large multimodal models in defending against autonomous vehicles attacks. *2024 IEEE Intelligent Vehicles Symposium (IV 2024)*, 3334–3341.

Li, J., Xu, R., **Ma, J.**, Zou, Q., Ma, J., & Yu, H. (2023). Domain adaptive object detection for autonomous driving under foggy weather. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2023)*, 612–622.

Ma, J., Pang, S., Yang, B., Zhu, J., & Li, Y. (2020). Spatial-content image search in complex scenes. *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2020)*, 2503–2511.

REFEREED WORKSHOP & SHORT PAPERS

Ma, J., Aldeen, M. S., Luo, F., & Cheng, L. (2025). Few-shot detection of hate videos using multi-modal large language models. *ACM Workshop on Deepfake, Deception, and Disinformation Security (3D-Sec '25)*, 32–35.

Han, X.^{*}, **Ma, J.**^{*}, Zhang, J., Liu, K., & Luo, F. (2025). Understanding the constraints of RAG-based medical LLMs: A case study in ophthalmic report generation. *2025 IEEE International Conference on Data Mining Workshops (ICDMW 2025)*, 96–102.

Aldeen, M. S., MohajerAnsari, P., **Ma, J.**, Chowdhury, M., Cheng, L., & Pesé, M. D. (2024). WIP: A first look at employing large multimodal models against autonomous vehicle attacks. *ISOC Symposium on Vehicle Security and Privacy (VehicleSec 2024)*.

Aldeen, M., Silimkhan, P., Anderson, E.[†], Kavuru, T.[†], Chang, T., **Ma, J.**, Luo, F., Hu, H., & Cheng, L. (2024). An integrated platform for online abuse research. *5th International Workshop on Cyber Social Threats (CySoc 2024)*.

Hatay, E., **Ma, J.**, Sun, H., Fang, J., Gao, Z., & Yu, H. (2021). Learning to detect phone-related pedestrian distracted behaviors with synthetic data. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2021)*.

Conference Presentations

Jan. 2026. Poster: *Understanding the Risks of Asphalt Art on the Reliability of Surveillance Perception Systems*. Transportation Research Board (TRB) Annual Meeting, Washington, D.C., USA.

Oct. 2025. Poster: *Potential Risks of Asphalt Arts on the Reliability of Perception Systems*. IEEE Secure Development Conference (SecDev 2025), Indianapolis, IN, USA.

Oct. 2023. Poster: *ICOAR: Integrative Cyberinfrastructure for Online Abuse Research*. IEEE Secure Development Conference (SecDev 2023), Atlanta, GA, USA.

Research Experience

Clemson University — Trustworthiness of AI

Clemson, SC, USA

ADVISOR: DR. LONG CHENG

Aug. 2023 – Present

- Investigated adversarial vulnerabilities in vision-based perception systems; developed diffusion-based defense architectures (*DisPatch*) against physical adversarial patches and quantified perception reliability under real-world visual perturbations.

Clemson University — AI for Online Content Safety

Clemson, SC, USA

ADVISOR: DR. LONG CHENG

Aug. 2023 – Present

- Formulated multimodal systems leveraging foundation models (LLMs/LVLMs) for few-shot hate video detection and automated community notes generation to counter contextual misinformation and facilitate online safety interventions.

Singapore University of Technology and Design — Biometric Authentication

Singapore

ADVISOR: DR. JIANYING ZHOU

May 2024 – Aug. 2024

- Developed multimodal biometric authentication architectures combining complementary physiological and behavioral signals and integrated cryptographic protocols to prevent spoofing and adversarial tampering.

Cleveland State University — Autonomous Vehicles & Object Detection

Cleveland, OH, USA

ADVISOR: DR. HONGKAI YU

Aug. 2021 – Jul. 2023

- Researched robust perception for autonomous vehicles under adverse weather and lossy V2V communication, and developed multimodal RGB-X fusion models for object-of-interest detection.

Brookhaven National Laboratory — Collision Avoidance Systems

Upton, NY / Cleveland, OH, USA

ADVISORS: DR. YONGHUA DU, DR. LU MA, DR. YUEWEI LIN, DR. HONGKAI YU

Mar. 2022 – Jul. 2023

- Engineered real-time computer vision pipelines for workspace collision avoidance in scientific beamline facilities, implementing spatio-temporal video object segmentation and multi-object tracking architectures.

Xi'an Jiaotong University — Image Retrieval

Xi'an, Shaanxi, China

ADVISOR: DR. SHANMIN PANG

Sep. 2016 – Jun. 2019

- Conducted research on semantic and spatial-content image retrieval, developing deep feature aggregation, similarity propagation, and heat diffusion graph re-ranking methods for complex scene queries.

Grant Proposal Participation

Contributed to the preparation and technical narrative development of the following grant proposals:

- **NCAE-C** — Towards Robust Perception Systems for Autonomous Vehicles Against Adversarial Attacks.
- **NSF FAIROS** — Transforming Social Science Research with an AI-Agent-Enabled Open Platform.
- **NSF ExLENT** — Exploring Generative AI Cybersecurity through Experiential Learning.
- **NSF SaTC** — Cognitive and Sociotechnical Safeguards for Protecting Young Adults from Unsafe AI-Generated Content.

Industry Experience

Chengdu Hsintiao Medical Technology Co., Ltd.

Chengdu, China

ALGORITHM ENGINEER

Oct. 2020 – Aug. 2021

- Developed deep learning algorithms for ECG signal analysis and automated clinical feature extraction.
- Optimized model training and inference pipelines for deployment on healthcare computing platforms.

Southeast Research Institute of China Unicom

Fuzhou, China

RESEARCH & DEVELOPMENT ENGINEER

Aug. 2019 – Sep. 2020

- Collected and annotated surveillance datasets under varied river conditions, training lightweight detection models.
- Built and deployed a computer vision system for real-time floating trash detection on rivers for environmental monitoring.

Teaching & Mentoring Experience

TEACHING EXPERIENCE

Clemson University

Clemson, SC, USA

TEACHING ASSISTANT

Aug. 2023 – Present

Assisted with course instruction across undergraduate and graduate levels; responsibilities included holding weekly office hours, grading coursework and exams, developing hands-on laboratory exercises, and delivering occasional guest lectures:

- CPSC 4200/6200 Computer Security Principles (Fall 2026)
- CPSC 4240/6240 System Administration and Security (Fall 2026)
- CPSC 3120 Introduction to Design and Analysis of Algorithms (Spring 2026)
- CPSC 8430 Deep Learning (Fall 2024, Spring 2025, and Fall 2025)
- CPSC 4300/6300 Applied Data Science (Spring 2024)
- CPSC 4030/6030 Computer Data Visualization (Fall 2023)

RESEARCH MENTORSHIP

Clemson University

Clemson, SC, USA

RESEARCH MENTOR

Jan. 2025 – Present

- **Arya Hemani & Janvi Patel** (Fall 2026): Mentoring two undergraduate students on generative UI/UX safety, guiding experimental design and automated auditing pipelines to evaluate deceptive dark patterns in prompt-to-design AI tools.
- **Addison Key** (Summer & Fall 2026): Mentoring an honors undergraduate researcher through Clemson's EUREKA! program on adversarial robustness benchmarking, threat modeling, and evaluation pipelines for vision-language-action (VLA) models.
- **Anvitaa Rudharraju** (Summer 2026): Mentored a high school student researcher on benchmarking vision-language models for ophthalmic tasks, including disease recognition and clinical diagnosis generation.
- **Ethan Anderson & Taran Kavuru** (Spring & Fall 2025): Mentored two undergraduate students on multimodal web safety, directing dataset curation and evaluation pipelines for research on online abuse and contextual misinformation.

Academic Service

2027	International AAAI Conference — AI for Social Impact Track (AISI),	Reviewer
2026	International AAAI Conference on Web and Social Media (ICWSM),	PC Member
2026	ACM International Conference on Multimedia (ACM MM),	PC Member
2025	Transportation Research Board Annual Meeting (TRB),	Reviewer
2025	IEEE Transactions on Dependable and Secure Computing (TDSC),	Reviewer
2025	The ACM Web Conference (WWW),	Reviewer
2025	IEEE International Conference on Distributed Computing Systems (ICDCS),	Subreviewer
2025	IEEE International Conference on Computer Software and Applications (COMPSAC),	Subreviewer
2025	ACM Advances in Social Networks Analysis and Mining (ASONAM),	Subreviewer
2024	ACM International Conference on Multimedia (ACM MM),	Reviewer
2023	ACM International Conference on Multimedia (ACM MM),	Reviewer
2022	ACM International Conference on Multimedia (ACM MM),	Reviewer

Honors and Awards

- 2024 Certificate of Appreciation (Volunteer), ACM AsiaCCS.
- 2024 Badge of Recognition, The ACM Web Conference.
- 2019 Excellent Graduation Thesis, Xi'an Jiaotong University.
- 2019 Excellent Graduate, Xi'an Jiaotong University.
- 2018 Successful Participant, National Graduate Mathematical Contest in Modeling (CN).
- 2018 Zhonghui Scholarship, Xi'an Jiaotong University.